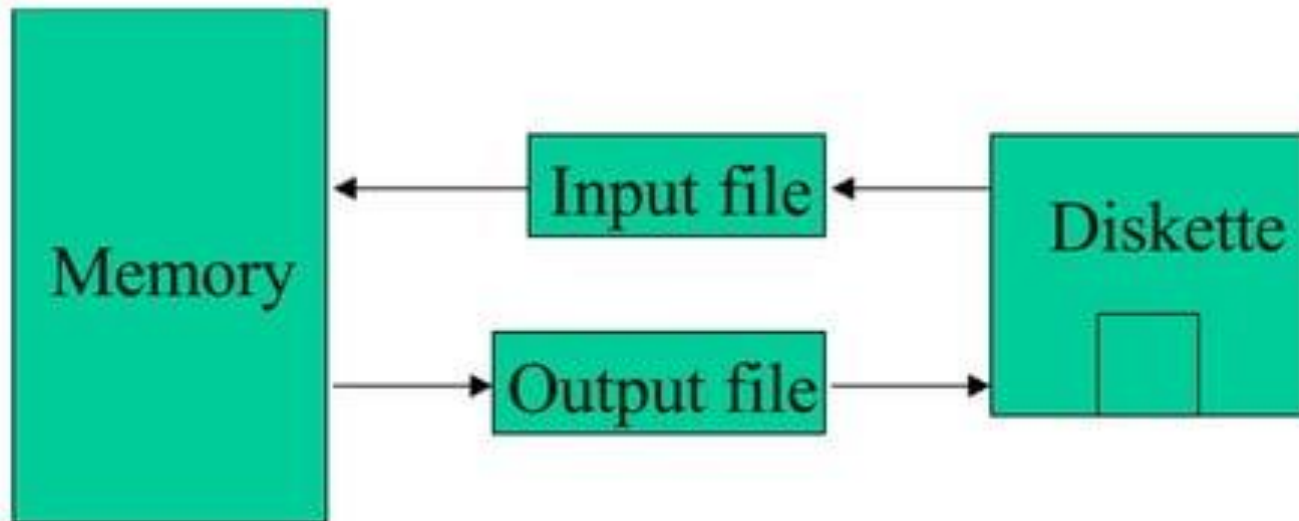


What is a File?

- A file is a collection of information, usually stored on a computer's disk. Information can be saved to files and then later reused.
- Computers store files to secondary storage so that the contents of files remain intact when a computer shuts down.
- When a computer reads a file, it copies the file from the storage device to memory; when it writes to a file, it transfers data from memory to the storage device.
- All files are assigned a name that is used for identification purposes by the operating system and the user.

Files

Used to transfer data to and from disk



Steps in Processing a File

Using a file in a program is a simple three-step process

- The file must be opened. If the file does not yet exist, opening it means creating it.
- Information is then saved to the file, read from the file, or both.
- When the program is finished using the file, the file must be closed.

File streams in c++

The following classes in c++ have access to file input and output functions.

- ifstream** – This data type represents the input file stream and is used to read information from files
- ofstream** – This data type represents the output file stream and is used to create files and to write information to files.
- fstream** – This data type represents the file stream generally, and has the capabilities of both ofstream and ifstream which means it can create files, write information to files, and read information from files.

Opening Files Using Constructor

ofstream - output file stream

- File access requires the inclusion of header file `fstream.h`
- Before data can be written to or read from a file, the file must be opened.
- Use ***ofstream*** for output stream.

ofstream objectname("filename");

E.g. // ofstream constructor opens file named "customer" for output(write).
fout as ofstream object.

```
void main()
{
    ofstream fout( "customer"); //open file to write
    string str_name="John";
    fout<<str_name;             //write to file
    .....
}
```

ifstream – input file stream

ifstream for input stream.

E.g. **Void main()**
 {

 ifstream fin("customer");
 fin>>string_name; //read from file

 }

Opening Files using open()

- Open multiple files using same stream object.

- **syntax:**

file-stream-class stream-object;

stream-object.open("filename",filemode);

Example:

```
ofstream fout;
```

```
fout.open("customer1");
```

```
.....
```

```
.....
```

```
fout.close();
```

```
fout.open("customer2");
```

```
.....
```

```
.....
```

```
fout.close();
```



optional

File Open Modes

ios:: in - (input) open a file to read

ios::out - (output) open a file to write

ios:: app - append to end-of-file.

ios: trunc -(truncate) delete the file contents if it exists.

ios:nocreate – open fails if the file doesn't exist.

ios:noreplace – open fails if file already exists.

fstream – input and output file stream

- **fstream** object can handle both the input and output(read and write) simultaneously.

```
void main()
{
    string name="John";
    fstream f; //created object f for fstream;
    f.open("TEXT", ios :: in | ios :: out); //opens file TEXT in read & write mode
    f<<name; // write name to file
    f>>name; // read name from file
    cout<<name;
}
```

Closing a File

- When we finish with a mode, we need to close the file before ending the program or beginning another mode with that same file.
- To close a file, we use `close` using the object:
`f.close () ;`